MIL-A-11210G
4 March 1981
SUPERSEDING
MIL-A-11210F
30 May 1973

MILITARY SPECIFICATION

AIR CONDITIONERS: BASE MOUNTED, SELF-CONTAINED, MULTIPACKAGE-TYPE, AIR-COOLED, ELECTRIC-MOTOR-DRIVEN, 18,000-BTU/HR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers self-contained base mounted air conditioners. The air conditioners control temperatures and humidities, and are suitable for use in extreme hot and humid conditions.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

BB-F-1421	- Fluorocarbon Refrigerants.
QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple Wall.

FSC 4120

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-116	- Preservation-Packaging, Methods of.
MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-E-52798	- Enamel, Alkyd, Camouflage.
MIL-E-52835	- Enamel, Modified Alkyd, Camouflage, Lusterless.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-186	- Protective Finishing Systems for Rockets, Guided Missiles, Support Equipment and Related Materials.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-882	- System Safety Programs for Systems and Associated Subsystems and Equipment, Requirements for.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, with Appropriate Test Methods.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.
MIL-STD-1474	- Noise Limits for Army Materiel.

DRAWINGS

ME

TA13217E0000	- Air Conditioner, Multipackage, 18,000 BTU/HR, 120/208 Volt, 50/60 Hertz, 3 Phase.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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SOUTHERN CALIFORNIA AIR POLLUTION CONTROL DISTRICT

Rule 102 - Photochemically Reactive Solvents.

(Application for copies should be addressed to the Southern California Air Pollution Control District, 434 S. San Pedro Street, Los Angeles, CA 90013).

AMERICAN SOCIETY OF HEATING, REFRIGERATING, AND AIR CONDITIONING ENGINEERS, INC. (ASHRAE)

Standard 37 - Methods of Testing for Rating Unitary Air Conditioning Equipment.

(Application for copies should be addressed to the American Society of Heating, Refrigerating, and Air Conditioning Engineers, Inc., 345 East 47th Street, New York, NY 10017.)

AMERICAN WELDING SOCIETY, INC. (AWS)

D1.1 - Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to the American Welding Society, Inc., 2501 NW 7th Street, Miami, FL 33125.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code - Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

3. REQUIREMENTS

3.1 Description. The air conditioners shall be in accordance with Top Assembly TA13217E0000 and as specified herein. The air conditioner shall operate on both 50 Hertz and 60 Hertz electrical power.

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The maximum weight of the air conditioner when charged with refrigerant and oil shall be 565 pounds.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissable without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the air conditioners. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request for inspection by the contracting officer or his designated representative.

3.2 First article (preproduction model). The contractor shall furnish one or more air conditioners as specified (see 6.2) for examination and testing within the time frame specified (see 6.2) to prove prior to starting production that his production methods will produce air conditioners that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4; and, unless otherwise specified herein, all examination and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the preproduction examination and tests, as specified (see 6.2).

3.3 Initial production. When specified (see 6.2), the contractor shall furnish to the Government one or more air conditioners (see 6.2) for inspection as specified in 4.4.

3.4 Material. Material shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.10).

3.5 Performance characteristics.

3.5.1 Capacity. When operating on 60 Hz electrical power and at the conditions as specified herein, the cooling capacity of the air conditioner shall be not less than as shown in table I.

TABLE I. Capacity rating conditions.

Condition	Ambient Temperature ° F	Evaporator air inlet temperature ° F		Total Net sensible cooling capacity	Total Net cooling capacity
	Dry bulb	Dry bulb	Wet bulb	Btu/Hr	Btu/hr
A	125	90	75	10,000	18,000
B	95	80	67	10,000	18,000

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3.5.2 Operating temperatures. The air conditioners shall be capable of functioning as follows:

(a) Start, operate, and cycle manually on cooling cycle at not less than plus 125° F ambient temperature with air at not less than plus 125° F entering the evaporator and condenser.
(b) Operate on cooling cycle without forming frost or ice on the evaporator at 60° F ambient temperature with air at plus 75° F dry bulb and plus 63° F wet bulb entering the evaporator.

3.5.3 Evaporator airflow. When the air conditioners are operating as specified in 3.5.1, the evaporator blowers shall have a combined capacity within the range of 800 to 1000 standard cubic feet per minute (scfm) to the space being cooled when operating against a static pressure of 0.25 inch of water, external to the air conditioner.

3.5.4 Power consumption. When the air conditioners are operating as specified in 3.5.1, the maximum total power consumption shall be not more than 3,000 watts. The condenser and evaporator blowers shall be wired to separate phases.

3.5.5 Pressures. When the air conditioners are operating as specified in 3.5.1, the condensing pressure shall be not more than 247 pounds per square inch gage (psig) and the suction pressure shall be not less than 40 psig.

3.6 Storage temperature. The air conditioners shall not be damaged and their operation shall not be impaired by storage at an ambient temperature of plus 155° F for 4 hours each 24 hours, no solar radiation, and at minus 65° F for 12 hours each 24 hours, no solar radiation.

3.7 Salt fog and humidity. When subjected to the tests specified in 4.6.3.12 and 4.6.3.13 the air conditioners shall show no evidence of deterioration, corrosion, or change in tolerance limits of any internal or external parts which could prevent the unit from meeting operational, service, or maintenance requirements, or shall show no evidence of peeling of paint, malfunctioning, refrigerant leaks, rough operation, damage, electrical shorts, erratic pressure or other irregular operation.

3.8 Rain. When subjected to the tests specified in 4.6.3.11, the air conditioners shall operate without functional damage or impairment of operation. There shall be no penetration of water from the evaporator section on the dry side of the enclosure or into or on any electrical component (such as motor, electrical connector, terminal switch, or control) that could result in an electrical short, cause damage from rust or corrosion, or become a hazard to personnel.

3.9 Safety. All exposed parts which are subject to high operating temperatures and pressure or which are energized electrically shall be insulated, fully enclosed, or guarded. All moving parts which are of such nature or so located as to be a hazard to operating or maintenance personnel shall be fully enclosed or guarded. Protective devices shall not impair the operating functions.

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3.10 Reliability. The specified Mean-Time-Between-Failures (MTBF) shall be 960 hours and the minimum acceptable MTBF shall be 480 hours when tested as specified in 4.6.3.17.

3.11 Refrigeration system. The refrigeration system shall be of the vapor cycle type employing refrigerant conforming to BB-F-1421, type 12 and will be furnished with a full charge in the system. The moisture content in the system shall be not more than 25 parts per million (ppm) by weight in the liquid phase. The entire refrigerant system shall withstand pressures up to 300 psig without deformation of any components or any detectable leak when tested as specified in 4.6.3.2.

3.12 Electrical controls. Electrical controls shall provide automatic operation of the air conditioner except for setting the controls on the control panel and resetting the protective devices following a malfunction. Electrical controls shall consist essentially of a selector switch, conditioner-air-thermostat, and a high pressure cut-out.

3.12.1 Selector switch. The selector switch shall control operation of the air conditioner as follows: When the switch is in the "OFF" position and power connected to the air conditioner, all components of the air conditioner shall be de-energized and inoperative. When the switch is in the "VENTILATION" position, the evaporator fan motor shall start and operate. When the switch is in the "COOLING" position, the evaporator fan motor shall continue to operate, and the operation of the condenser fan motor and compressor motor shall be accomplished as demanded by the conditioned-air-thermostat. When the switch is in the "EXTERNAL CONTROL" position, the cooling operation of the air conditioner shall be controlled by an external thermostat.

3.12.2 Conditioned-air-thermostat. The conditioned-air-thermostat shall control the cooling temperature automatically within 2-1/2° F of any preset temperature between 60° F and 90° F by cyclic operation of the air conditioner.

3.12.3 High pressure cutout switch. The high pressure cutout switch shall be set to de-energized the compression and condenser fan motors when the condensing pressure reaches 280 ± 7 psig. The switch shall require manual resetting after being tripped.

3.13 Control panel. The control panel shall be capable of being remotely located from the air conditioner and mountable on the end panels of the evaporator blower section without alterations other than extension of the wiring. The air conditioner shall be furnished with the control panel located in the condenser blower section. When specified (see 6.2), the panel shall be mounted on an end panel of the evaporator blower section as specified (see 6.2).

3.14 Motors. All motors shall be equipped with automatic-resetting thermal overload protectors sized to prevent motor operation above the safe operating temperature for which the motor is rated.

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3.15 Voltage connections. The air conditioners shall be furnished wired for either 120 volts or 208 volts electrical power supply as specified (see 6.2). Current sensing and overload-trip devices for the alternate voltage shall be provided, bagged, identified, and tied to the air conditioner control panel selector switch. Instructions for converting the air conditioner for operation on the alternate voltage shall be packaged with the trip devices.

3.16 Wiring. Wiring shall be secured in compact harnesses and attached to frame members with insulated clamps at close intervals to insure a neat and orderly cable run. Cables and wires shall not be spliced at any point throughout the length of their runs. Wiring shall not be attached to panels or other components that require removal during maintenance, except for actual electrical connections to components. All wiring between sections shall pass through the same opening which shall be grommeted. Electrical isolators shall be used between all wiring and metal components to prevent abrasive action to wire insulation.

3.17 Electromagnetic interference. The electromagnetic interference emission characteristics of the air conditioners shall conform to MIL-STD-461A, Notice 1, table A-I of the appendix, when tested as specified in 4.6.3.16 (see 6.8).

3.18 Fungus and moisture resistance. The electrical circuitry, including all components and connections except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, composition I or II as applicable (see 3.18.1), with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish:

(a) Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.18.1 Composition. Composition II shall be used only in the case when local air pollution regulations governing the application of varnish precludes the use of composition I. When composition II is used, the contractor shall provide evidence to the Government that the use of composition II is required, and shall certify that the composition II material complies with Rule 102 - Southern California Air Pollution Control District.

3.19 Treatment and painting. Unless otherwise shown on the drawings, all parts shall be cleaned, treated, and painted in accordance with MIL-STD-186 with the following exceptions:

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(a) Internal components previously painted by the manufacturer need not be repainted if they conform to the salt-fog requirements.
(b) Coils, refrigerant tubing, electrical components, or other components whose operation will be impaired by painting shall not be painted.
(c) Dissimilar metals shall be insulated from one another in accordance with MIL-T-704.

Unless otherwise specified (see 6.2 and 6.9), the color of the finish coat shall be forest green in accordance with MIL-E-52798 or MIL-E-52835.

3.20 Instruction plates. Each air conditioner shall be equipped with instruction plates or diagrams, including warnings and cautions, describing any special or important procedures to be followed in assembling, operating or servicing the air conditioner. A schematic wiring diagram of the air conditioner shall be attached to the front panel of the condenser blower section. The diagram shall include instructions for converting the air conditioner from one voltage to the other. A schematic refrigeration diagram shall also be attached to the front of the condenser blower section. The diagram shall show all components and lines in the refrigeration system with arrows indicating direction of refrigerant flow when the air conditioner is operating on the cooling cycle. The schematic refrigeration diagram shall also show the type and quantity of oil and refrigerant required by the system. Instruction plates shall conform to MIL-P-514, type III, composition C, of type I, grade A, class 1 material. Plates shall be attached by screws, bolts, or rivets.

3.21 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more parts possessing such functional and physical characteristics as to be equivalent in performance and durability and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts except for adjustment, and without selection for fit or performance.

3.22 Maintenance. The availability shall be not less than 95 percent. The Mean-Time-To-Repair (MTTR) for the operating crew and organizational maintenance shall be not more than 4 hours, for direct support shall be not more than 8 hours and for general support shall be not more than 24 hours. The Mean-Time-Between-Maintenance (MTBM) for the operating crew and organizational maintenance shall be not less than 250 hours, for direct support shall be not less than 1000 hours, and not less than 4000 hours for general support.

3.23 Noise level. The noise level in a conditioned area at a properly engineered installation shall conform to MIL-STD-1474, table II, category E. The condenser side noise level shall conform to MIL-STD-1474, table II, category D.

3.24 Workmanship.

3.24.1 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that

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will not cause injury to the material. Corners shall be square and true. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.24.2 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or cap-screws shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement.

3.24.3 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes and in full contact with the surface of the member.

3.24.4 Jigs and fixtures. Shop-fabricated components shall be assembled in steel jigs or frames and joined while held in position. Jigs and frames shall be designed to minimize distortion.

3.24.5 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operation to be performed and that such qualification is effective as defined by the particular code:

AWS D1.1 - Structural Welding Code, Section 5, Qualification.
ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

Contractors who only make horizontal welds need not qualify welders for "all position welding". Subject to approval by the Government, contractor's standard welder qualification may be substituted in lieu of the above codes provided that the contractor's procedure is equivalent to the above codes. The contractor shall be responsible for determining that automatic welder equipment operators are capable of producing quality welds.

3.24.6 Brazing. During the brazing operation a small amount of gas such as dry nitrogen, shall be continuously bled through the refrigerant lines to prevent internal oxide scaling. Care shall be exercised during the brazing operation to protect all components from deformation and damage. The quantity of brazing alloy and flux used shall be no more than that required to make the joint. Flux shall not be present in the refrigerant system and heavy beads and fillets of braze material shall be avoided.

4. QUALITY ASSURANCE PROVISONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection

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requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Acceptability criteria. Air conditioners which conform to all requirements of Section 3 and 5 of this specification and pass all applicable examinations and tests in Section 4 of this specification will be considered acceptable by the Government.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.1.3 Disassembly inspection. Failure of any test or examination of the preproduction model shall be cause for disassembly, in the presence of a Government representative, of the preproduction model to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejecting the preproduction model.

4.1.4 End item final inspection requirement (EIFIR). Each air conditioner shall be inspected in accordance with the applicable EIFIR. One copy of the completed EIFIR shall be shipped with each inspected air conditoner (see 6.4).

4.1.5 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the Supplementary Quality Assurance Provisions (SQAP) shown on the drawings. The drawings specify the characteristics requiring SQAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.4 and 6.5).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of packaging (see 4.8).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction air conditioners shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection or cause for performing the inspection specified in 4.1.3 as applicable.

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4.3.2 Tests. The preproduction air conditioner shall be subject to the tests marked "X" in column 2 of table III in the sequence listed or in the sequence listed in a Government approved test plan. Failure of any test shall be cause for performing the inspection specified in 4.1.3.

4.4 Initial production inspection. When specified (see 3.3), one or more initial production air conditioners will be selected at random by the Government from the air conditioners being produced by production tooling and will be subject to the examination and tests marked "X" in column 4 of table III to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production air conditioner shall not exclude the remaining air conditioners from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any and all tests contained in this specification as part of the initial production test and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.4.1 Inspection failure. Failure of an initial production air conditoner to meet any requirement specified herein during and as a result of the inspection specified in 4.4 shall be cause for rejection of the initial production air conditioners and shall be cause for refusal by the Government to continue acceptance of production air conditioners until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on air conditioners previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all air conditioners accepted prior to the completion of initial production inspection are similary deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Examination. Each air conditioner shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.5.2 Tests. Each air conditioner shall be subject to the tests marked "X" in column 3 of table III in the sequence listed. Failure of any test shall be cause for rejection.

4.6 Inspection procedure.

4.6.1 Examination. Each air conditioner shall be examined for the characteristics listed in table II as specified in 4.5.1.

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TABLE II. Examination schedule.

Number	Characteristics	Requirement Paragraph
Major		
101.	Nonconformance to the drawings.	3.1.1
102.	Material not as specified.	3.4
103.	Electrical controls not as specified.	3.12
104.	Selector switch not as specified.	3.12.1
105.	Control panel not as specified.	3.13
106.	Motors not equipped with automatic-resetting thermal overload protectors sized as specified.	3.14
107.	Voltage connections not as specified.	3.15
108.	Wiring not as specified.	3.16
109.	Fungus and moisture resistance protection not as specified.	3.18, 3.18.1
110.	Instruction plates not as specified.	3.20
111.	Interchangeability not as specified.	3.21
112.	Metal fabrication not as specified.	3.24.1
113.	Bolted connections not as specified.	3.24.2
114.	Riveted connections not as specified.	3.24.3
115.	Jigs and fixtures not as specified.	3.24.4
116.	Welders not certified as specified.	3.24.5
117.	Brazing not as specified.	3.24.6
118.	Weight of the air conditioner not as specified.	3.1
119.	Current sensing and overload-trip devices for the alternate voltage missing.	3.15
120.	Electromagnetic interference suppression components and network not identical with approved model.	3.17
Minor		
201.	Treatment and painting not as specified.	3.19
202.	Workmanship not as specified.	3.24

4.6.2 Test schedule. The inspection test schedule is contained in table III.

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TABLE III. Test schedule 1/

Type of Test				Name of test	Inspection Paragraph	Requirement Paragraph
Insp. Seq.	Preprod.	Qual. Conf.	Init. Prod and Insp. Comp.			
1	2	3	4	5	6	7
1	X	X	X	Examination.	4.6.1	3.1, 3.4, 3.13, 3.15, 3.16, 3.18, 3.19, 3.20, and 3.25
2	X	X	X	Refrigerant-system pressure test.	4.6.3.2.1	3.11
3	X	X	X	Controls test.	4.6.3.3	3.12
4	X	X 2/	X	Refrigerant moisture content test.	4.6.3.4	3.11
5	X	-	X	Electromagnetic interference.	4.6.3.16	3.17
6	X	X 3/	X	Capacity and air flow test.	4.6.3.5	3.5.1, 3.5.3, 3.5.4, 3.5.5
7	X	-	X	Cycling test.	4.6.3.6	3.5.2
8	X	-	X	Evaporator coil frost.	4.6.3.7	3.5.2
9	X	-	X	High temperature operation.	4.6.3.8	3.5.2
10	X	-	X	High temperature storage.	4.6.3.9	3.6
11	X	-	X	Low temperature storage.	4.6.3.10	3.6
12	X	-	X	Rain.	4.6.3.11	3.8
13	X	-	X	Humidity.	4.6.3.12	3.7
14	X	-	X	Salt fog.	4.6.3.13	3.7
15	X	-	X	Controls.	4.6.3.3	3.12
16	X	-	X	Motor protection.	4.6.3.14	3.14
17	X	X	-	Production run-in.	4.6.3.15	3.10
18	-	-	X	Refrigeration-leakage.	4.6.3.2.2	3.11
19	-	-	X	Reliabilty (MTBF).	4.6.3.17	3.10
20	-	-	X	Maintenance evaluation.	4.6.3.18	3.22
21	-	-	X	Noise level test.	4.6.3.19	3.23
22	-	-	X	Safety.	4.6.3.20	3.9

1/ Perform inspection marked "X".
2/ Test performed on first production unit. Contractor may certify remaining production units if the method for evacuating, dehydrating, and charging of refrigerant system is identical to first production unit.

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3/ Test performed only if the production air conditioner fails test specified in 4.6.3.15.

4.6.3 Tests.

4.6.3.1 Conditions. Unless otherwise specified herein, tests shall be conducted at rated voltage and employing 60 Hz electrical power. The frequency tolerance for all tests shall be plus or minus 1 percent. Components subject to atmospheric corrosion shall not be treated with any treatment such as grease or oil in order to meet test requirements. Capacity and airflow tests shall be conducted utilizing the instrument, apparatus, and procedures specified in ASHRAE Standard 37. Production run-in tests shall be conducted as specified herein utilizing instruments specified in ASHRAE Standard 37 and as specified herein. The test chambers shall be located within a controlled environment so that the chamber is not affected by sudden changes in outside climate conditions during a test run.

4.6.3.2 Refrigerant-system pressure and leakage.

4.6.3.2.1 Refrigerant-system pressure. Test pressure tolerances shall be plus 0 and minus 5 psig. Charge the refrigerant system to 300 psig with dry nitrogen containing refrigerant gas detectable with a halogen-sensitive electronic leak detector (General Electric type H-2, or equal). Calibrate the leak detector for the type of gas and percentage of mixture employed and set the detector to indicate a leakage rate of 0.1 ounce per year. Leak test the entire refrigerant system for leaks. One or more of the following shall constitute failure of the tests:

(a) Deformation of any component in the refrigerant system.
(b) Any refrigerant leak from the refrigerant system of 0.1 ounce per year or more.

4.6.3.3 Controls. Conduct the controls test at an ambient temperature between plus 65° F and 85° F. Ambient temperature shall be determined adjacent to the air conditioner thermostat.

4.6.3.3.1 Selector switch. Set the selector switch on the "COOL" position. Determine that the evaporator blowers operate continuously and operation of the compressor and condenser blowers are controlled by the thermostat. Set the selector switch in the "OFF" position. Determine that no component runs at any setting of the thermostat. Set the selector switch in the "VENT" position. Determine that the evaporator blowers run at any setting of the thermostat. Set the selector switch in the "EXTERNAL CONTROL" position. Determine that the evaporator blowers operate continously and operation of compressor and condenser blowers are controlled by an external thermostat. An external thermostat is not required with the air conditioner. Any external thermostat of the proper voltage may be used by the contractor to demonstrate operation on "EXTERNAL CONTROL" position. Nonconformance to 3.12.1 shall constitute failure of this test.

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4.6.3.3.2 Thermostat. Set selector switch in "COOL" position. Set thermostat below ambient temperature. Determine that the air conditioner is operating on cooling cycle with evaporator fan, compressor, and condenser fan running. Allow room temperature to drop to thermostat setting. Determine that the compressor and condenser fan stops, and the evaporator fan continues to run. Allow room temperature to raise 2-1/2° F. Determine that the air conditioner starts and operates on cooling cycle. Inability of the thermostat to control the air conditioner as specified in 3.12.2 shall constitute failure of this test.

4.6.3.3.3 High-pressure cutout controls. Operate the air conditoner on the cooling cycle with pressure gages installed on the compressor service valves. Raise the condensing pressure by reducing the condenser airflow. Determine at what pressure the compressor motor and condenser fan motor is de-energized and inoperative. Remove the condenser air blockage and reset the cutout control. Determine that the compressor motor and condenser fan motor restarts. Nonconformance to 3.12.3 shall constitute failure of this test.

4.6.3.4 Refrigerant moisture content. After not less than 1 hour of continuous operation in the cooling mode in an ambient temperature of not less than plus 95° F, shut down the air conditioner. After an additional period of 1 hour, withdraw a sample of refrigerant vapor from the low-pressure side and determine the moisture content by one of the methods specified in BB-F-1421. A moisture content of more than 25 ppm shall constitute failure of this test.

4.6.3.5 Capacity and airflow. The air conditioner shall be tested for cooling capacity and evaporator airflow in accordance with ASHRAE Standard 37, except that the two capacity rating conditions specified in 3.5.1 and the static pressure as specified in 3.5.3 shall be utilized for the capacity and airflow test. The capacity and airflow test shall be conducted simultaneously on the same air conditioner. The test arrangement shall be the room air-enthalpy test method for both evaporator and condenser side measurements. Separate the evaporator and condenser sections of the air conditioner and install a bottom panel on the evaporator section for the cooling capacity tests. The evaporator section shall be located completely in the indoor side test room, and the condenser section shall be located completely in the outdoor side test room. Install refrigerant liquid and suction lines through the center wall partition to complete the refrigerant system for the capacity tests. After the tests are completed remove the bottom panel from the evaporator section and reassemble the evaporator and condenser sections. Any of the following shall constitute failure of this test:

(a) A capacity of less than 18,000 Btu/hr with an airflow of 800 to 1000 cfm against a static pressure of 0.25 inch of water from both climatic conditions specified.
(b) Inability to maintain a minimum sensible cooling capacity of 10,000 Btu/hr.
(c) Condensing pressure more than 247 psig or suction pressure less than 40 psig.
(d) A total power consumption of more than 3,000 watts.

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4.6.3.6 Cycling. While each set of climatic conditions specified in 3.5.1 is maintained, cycle the air conditoner on and off not less than three times. Operate for not less than 15 minutes before turning the air conditoner off. After the air conditioner is turned off, allow a maximum of 3 minutes before restarting. Inability of the air conditioner to restart and to operate after the 3-minute waiting period shall constitute failure of this test. Repeat this test using 50 Hz power.

4.6.3.7 Evaporator-coil frost. Operate the air conditioner continuously for 6 hours on the cooling cycle at an ambient temperature of plus 60° F with air at plus 75° F dry bulb and plus 63° F wet bulb entering the evaporator. Any evidence of ice or frost on the evaporator coil shall constitute failure of this test.

4.6.3.8 High temperature operation. Subject the air conditioner to a temperature of plus 125° F plus 3° F minus 0° F for a period of 4 hours. After completion of the exposure period, operate the air conditioner in the cooling mode for a period of not less than 1 hour in an ambient temperature of plus 125° F plus 3° F minus 0° F with air at plus 125° F plus 3° F minus 0° F entering the evaporator and condenser. Turn the air conditioner "OFF" and then restart and operate the air conditioner in the cooling mode for a period of not less than 15 minutes. Repeat the "OFF" and 15 minute operating period two more times. Inability of the air conditioner to start, operate, restart, or nonconfomance to 3.5.2, shall constitute failure of this test.

4.6.3.9 High temperature storage. Subject the air conditoner to a temperature of plus 155° F for a period of 4 hours. At the conclusion of the exposure period, remove the air conditioner from the test chamber and examined for deterioration, corrosion, or damage. After completion of the examination, allow the air conditoner to soak at room ambient temperature for a period of not less than 1 hour but not more than 3 hours. After completion of the soaking period, operate the air conditioner on the cooling mode for a period of not less than 1 hour. Inability of the air conditioner to start, operate, or any evidence of deterioration, corrosion, damage, such as rupture of tubing, damage to control components, binding of motors, fans, or other mechanical parts, deterioration of gaskets, damage to insulation, breakdown of adhesives, permanent setting of gaskets or packing, or loss of refrigerant shall constitute failure of this test.

4.6.3.10 Low temperature storage. Subject the air conditioner to a temperature of minus 65° F for a continuous period of 12 hours. At the conclusion of the exposure period, remove the air conditioner from the test chamber and examine for deterioration and damage. After completion of the examination, connect power to the air conditioner and allow the air conditioner to soak at room ambient temperature for a period of not less than 6 hours. After completion of the soaking period, operate the air conditioner on cool mode for a period of not less than 1 hour. Any evidence of deterioration, permanent deformation, damage, or failure of the air conditioner to start and operate shall constitute failure of this test.

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4.6.3.11 Rain. Conduct the rain test with the air conditioner set up to simulate conditions of service with the air conditioner located outside of an enclosure. Evaporator supply and return air openings shall be fitted with ducts to prevent water entering and to allow normal airflow. The external duct system shall not impose a resistance to airflow in excess of 0.25 inch W.G. static pressure. The control panel shall be protected from the rain. Rainfall shall be produced by means of water spray and shall be measured with US Weather Bureau type gages. Nozzles shall emit droplets having a diameter of not less than 1.5 millimeters. The water supply temperature shall be between 50° F and 70° F and shall contain fluorescent dye for ease in determining the locations and extent of leakage. After each rain test, remove panels and component covers from the air conditioner as required to examine for water penetration. After the examination, operate the air conditioner for 15 minutes in the cooling mode.

4.6.3.11.1 Free rain. With the air conditioner not operating, subject each of the four sides and top of the air conditioner to a simulated rainfall of 4 inches per hour plus or minus 1 inch for 30 minutes each, for a total test duration of not less than 2-1/2 hours. Direction of rainfall against each of the three sides shall be varied throughout the test from vertical to 45 degrees from vertical.

4.6.3.11.2 Blowing rain. With the air conditioner operating on cool mode, subject each of the four sides to a simulated rainfall of 1 inch per hour plus or minus 1/2 inch combined with a horizontal wind of 40 miles per hour plus or minus 5 miles per hour for a period of 1 hour. Total test duration shall be not less than 4 hours.

4.6.3.11.3 Basis of failure. Nonconformance to 3.8 shall constitute failure of the rain test.

4.6.3.12 Humidity. Subject the air conditioners to test method 507, procedure I, of MIL-STD-810, except the maximum temperature shall be 155° F in lieu of 160° F. Do not operate the air conditioner during exposure and do not expose the air conditioner to extreme temperature prior to test. Prior to this test and after the exposure period, operate the air conditoner in accordance with 4.6.3.3 to determine function of the controls. Nonconformance to 3.7 shall constitute failure of this test.

4.6.3.13 Salt fog. Subject the air conditioners to test method 509 of MIL-STD-810. Do not operate the air conditioner during exposure. Prior to this test and after the exposure period, operate the air conditioner in accordance with 4.6.3.3 to determine function of the controls. Nonconformance to 3.7 shall constitute failure of this test.

4.6.3.14 Motor protection.

4.6.3.14.1 Measurements. Insulation resistance, motor coil winding resistance, motor winding temperature and motor electric current shall be measured for this test. Insulation resistance between the motor windings and fame of the motor shall be measured with a megger insulation tester with a

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maxmium voltage of 500 VDC. Winding resistance shall be measured with a wheatstone bridge. Fan motor winding temperatures shall be measured by means of thermocouples, consisting of wires not larger than No. 24 AWG, placed on surfaces of all windings integrally applied to the insulation on the conductors. Motor electric current shall be measured with a "clamp on" type ammeter.

4.6.3.14.2 Evaporator fan motor. Lock the fan motor shaft to prevent rotation. Measure the insulation resistance, motor winding resistance and motor winding temperature with the thermocouples installed on the windings. The motor winding temperature shall be 70° F plus or minus 10° F before beginning test. Place the air conditioner selection switch in the "OFF" position and apply electric power at the rated voltage and frequency to the air conditioner in the normal manner. Place the selector switch to "VENTILATION" position, energizing the fan motor with the shaft locked. Allow the motor to be cycled "on and off" for 5 hours by the automatic reset thermal protector installed on the motor. Measure and record the locked rotor current and the highest winding temperature during the third "on" cycle and at the end of each hour thereafter. At the completion of the test, de-energized the motor and allow motor winding temperature to equalize until a temperature of 70° F plus or minus 10° F is reached. Measure the motor winding resistance and insulation resistance. Remove the locking device from the fan motor shaft and operate the air conditioner in the cooling mode for not less than 1 hour. Record the fan motor running current at the end of the hour.

4.6.3.14.3 Condenser fan motor. Lock the fan motor shaft to prevent rotation. Measure the insulation resistance, motor winding resistance, and motor winding temperature with the thermocouples installed on the windings. The motor winding temperature shall be 70° F plus or minus 10° F before beginning test. Electrically disconnect the compressor and connect a jumper wire across the terminals of the thermostat. Place the air conditioner selector switch in the "OFF" position and apply electric power at the rated voltage and frequency to the air conditioner in the normal manner. Place the selector switch to "COOLING" position, energizing the condenser fan motor with the shaft locked (the evaporator fan will operate normally). Allow the condenser motor to be cycled "on and off" for 5 hours by the automatic reset thermal protector installed on the motor. Measure the locked rotor current and the highest winding temperature during the third "on" cycle and at the end of each hour thereafter. At the completion of the test, de-energize the motor and allow motor winding temperature to equalize until a temperature of 70° F plus or minus 10° F is reached. Measure the motor winding resistance and insulation resistance. Remove the locking device from the fan motor shaft, reconnect the compressor and operate the air conditoner in the cooling mode for not less than 1 hour. Record the condenser fan motor running current at the end of the hour.

4.6.3.14.4 Failure criteria for fan motors. One or more of the following shall constitute failure of this test:

(a) A motor winding resistant varying more than plus or minus 15 percent from the initial motor winding resistance.
(b) An insulation resistance of less than 150,000 ohms.

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(c) Failure of the air conditioner to operate during the motor protector test or during the production run-in test (see 4.6.3.15).
(d) A motor winding temperature exceeding the maximum recommendation of the motor manufacturer.
(e) Any evidence of smoke (visual) from the air conditioner during the motor protector test or during the production run-in test.
(f) Motor running current 20 percent above name plate amperage.
(g) Locked rotor current varying by 25 percent from the manufacturers furnished data.

4.6.3.14.5 Compressor motor. Place the air conditioner in an ambient temperature of 77° F plus or minus 1/2° F (25° C) for 24 hours. Measure and record the ambient temperature and the resistance of each phase winding of the motor. Place jumper wires around the circuit breaker to prevent de-energizing the power supply and disconnect power to the condenser fan motor. Immerse the conditoned air thermostat bulb in water at a temperature of not more than 35° F. Position the mode selector switch in the "COOL" position and the conditioned air thermostat clockwise to the maximum "COOLER" position. Operate the air conditioner until the thermal overload protection device for the compressor motor windings is activated. Measure the resistance of the compressor motor windings. Calculate the temperature of the windings as follows:

$$t = \frac{259.5R_2 - 234.5R_1}{R_1}$$

Where:
R_1 = resistance of windings at 25° C ambient.
R_2 = resistance of windings at time of acutation of thermal overload device.
t = temperature of motor windings ° C.

Check electrical continuity from phase to phase. Remove the jumper wires from the circuit breaker, reconnect power to the condenser fan motor, reinstall the conditioned air thermostat bulb, and operate the air conditioner in the cooling mode for 1 hour in an ambient temperature range of 80° F to 90° F. One or more of the following shall constitute failure of this test:

(a) Motor winding temperature (t) in excess of motor manufacturer's recommended limit.
(b) Lack of electrical continuity from phase to phase.
(c) A motor winding resistance varying more than plus or minus 15 percent from the initial motor winding resistance.
(d) An insulation resistance of less than 150,000 ohms.
(e) Failure of the air conditioner to operate during the motor protector test or during the production run-in test (see 4.6.3.15).
(f) Any evidence of smoke (visual) from the air conditoner during the motor protector test or during the production run-in test.
(g) Motor running current 20 percent above name plate amperage.

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4.6.3.15 Production run-in. The preproduction model which has been subjected to capacity and airflow tests shall be used to establish the criteria for testing all production air conditioners. The preproduction model shall be located immediately adjacent to and operated concurrently with the production unit from a common power source. Air condtioners shall be within an enclosure maintained at 90° plus or minus 2° F dry bulb temperature. The evaporator air discharge and inlet opening shall be equipped with sheet-metal ducts approximately 2 feet long. The evaporator air discharge opening shall employ a common duct. The discharge and inlet ducts shall each taper to an opening size as required to cause an air velocity of 700 to 1,000 feet per minute at the end of the duct. Each duct shall be equipped with two dry-bulb thermometers located 6 inches from the end of the duct away from the air conditioner and spaced over the cross section of the duct so as to give average entering and leaving air temperatures. Condensing pressure and suction pressure gages shall be attached to the units. Means shall be provided to check the power consumption in watts of each air conditioner. Operate each production air conditioner for not less than 1 hour under enclosure conditions. The following performance data shall be obtained during the last 2 minutes of this 1-hour period:

(a) Dry-bulb temperature entering the air conditioner, ° F, two thermometers.
(b) Dry-bulb temperature leaving the air conditioner, ° F, two thermometers.
(c) Condensing pressure, psig.
(d) Suction pressure, psig.
(e) Power consumption of the air conditioner, watts, one reading.

At the completion of the above, operate the air conditioner for 48 hours in the cooling mode in an ambient temperature between 100° F and 120° F. Upon completion of the 48-hour operation, repeat the performance test conducted prior to the 48-hour test and check the entire system for leaks with the detector specified in 4.6.3.2.2 calibrated for the gas employed. The above results obtained on each production air conditioner should not be at variance with the results on the preproduction model by more than the following:

(1) Temperature drop between the average entering dry-bulb temperature and the average leaving dry-bulb temperature more than 1° F less than preproduction model.
(2) Condensing pressure at variance from preproduction model by more than plus or minus 5 psig.
(3) Suction pressure at variance from preproduction model by more than plus or minus 2 psig.
(4) Total watts power consumption exceeds preproduction model by more than 3 percent.

One or more of the following shall constitute failure of this test:

(a) Failure of any component, fracture, or breakdown of any material.
(b) Refrigerant leakage in excess of 0.1 ounce per year except for compressor shaft seal. Refrigerant leakage at the shaft seal in excess of 1 ounce per year.

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(c) Variations from the preproduction valves greater than those specified above.

4.6.3.16 Electromagnetic inteference. The preproduction model shall be tested in accordance with Test Methods RE02 (0.15 MHz to 1,000 MHz) and A-CE1 (1.5 MHz to 65 MHz) as shown in table A-I of MIL-STD-461A, notice 1, appendix. The contractor shall furnish the contracting officer a test plan and test report as required by MIL-STD-461. Disapproval of the test report shall constitute failure of this test.

4.6.3.17 Reliability. Using the MTBF specified in 3.10, the air conditioners shall be tested as specified in 4.6.3.17.1 through 4.6.3.17.3 with "accept" and "reject" criteria in accordance with figure 2. Continue testing until an "accept" or "reject" decision is reached.

4.6.3.17.1 Test facility. Install the air conditioners in a test chamber consisting of two rooms. One room shall function as an indoor room and shall maintain the evaporator return air temperature as specified in figure 1 within plus or minus 5° F. The second room shall function as an outdoor room and shall maintain the condenser return air temperature as specified in figure 1 within plus or minus 5° F. These temperatures shall be maintained on a 24-hour cyclic basis as shown in figure 1 throughout the MTBF test.

4.6.3.17.2 Operation. All air conditioner controls (rotary switch and thermostat) shall be operated manually or automatically. If automatic means are used, they shall not interrupt the control circuit but shall physically operate the air conditioner control components. The controls shall be positioned as shown in table IV.

TABLE IV. Control positions for reliability test.

Mode	Time	Control	Control Position
Ventilate	0030 to 0200	Rotary Switch	Ventilate
Cooling	0200 to 0600	Rotary Switch	Cooling
		Thermostat	Extreme Clockwise
Cooling	0600 to 0630	Rotary Switch	Cooling
		Thermostat	Extreme Counter-clockwise
Cooling	0630 to 1000	Rotary Switch	Cooling
		Thermostat	Extreme Clockwise
Cooling	1000 to 1030	Rotary Switch	Cooling
		Thermostat	Extreme Counter-clockwise
Cooling	1030 to 1400	Rotary Switch	Cooling
		Thermostat	Extreme Clockwise
Cooling	1400 to 1430	Rotary Switch	Cooling
		Thermostat	Extreme Counter-clockwise

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TABLE IV. Control positions for reliability test (cont'd).

Mode	Time	Control	Control Position
Cooling	1430 to 1700	Rotary Switch Thermostat	Cooling Extreme Clockwise
Cooling	1700 to 1730	Rotary Switch Thermostat	Cooling Extreme Counter-clockwise
Cooling	1730 to 2000	Rotary Switch Thermostat	Cooling Extreme Clockwise
Cooling	2000 to 2030	Rotary Switch Thermostat	Cooling Extreme Counter-clockwise
Cooling	2030 to 2300	Rotary Switch Thermostat	Cooling Extreme Clockwise
Ventilate	2300 to 2330	Rotary Switch	Ventilate
Off	2330 to 0030	Rotary Switch	Off

4.6.3.17.3 Data. The following shall be monitored and recorded throughout the test to determine that normal air conditioner operation is being maintained and to identify problems in event of failure or malfunction:

(a) Voltage.
(b) Current.
(c) Frequency.
(d) Refrigerant pressure.
    (1) Suction.
    (2) Discharge.
(e) Refrigerant temperatures.
    (1) Compressor discharge.
    (2) Liquid line.
    (3) Suction line at compressor.
(f) Air temperature.
    (1) Evaporator return (db).
    (2) Evaporator discharge (db).
    (3) Condenser return (db).
(g) Abnormal operation (visual and audio observations).
(h) Sightglass observation (clear or bubbles).
(i) Identification of unit.
(j) Initials of technician.
(k) All failures.
(l) Time of day of failure.
(m) Action taken if failure occurs.
(n) Number of operating hours on unit.

4.6.3.17.4 Failure criteria. A failure is defined as any malfunction for which corrective action cannot be deferred until the next scheduled organizational preventive maintenance check. Corrective action is not deferrable if the

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malfunction causes or may cause inability to commence operation of the unit, cessation of operation of the unit, or degradation of performance capability of the air conditioner below the levels required for the accomplishment of its mission functions. Malfunctions will not be counted as "failures" for reliability purposes if detected and corrected during initial inspection. Malfunctions resulting from operational or maintenance errors will not be counted. The following are considered to be malfunctions:

(a) Failure to commence operation, cessation of operation, or degradation of performance below designated levels.
(b) Serious damage to the system or subsystem.
(c) Serious personnel safety hazards.

4.6.3.18 Maintenance evaluation. Maintenance operations required during preproduction and initial production tests shall be accomplished as specified in 3.22. Any assemblage or support equipment such as technical manuals, repair parts, special tools, etc., provided under the contract for operation or conducting maintenance will be evaluated. Nonconformance to 3.22 shall constitute failure of this test.

4.6.3.19 Noise level. To determine the noise level of the air conditioner at the evaporator and condenser sections, place three microphones on an arc 46 inches from the center of each section in the same horizontal plane; record the noise level while the air conditioner is in normal operation with the compressor running. In each section, one microphone shall be placed directly in line with the center of the air conditioner; at the evaporator section, place one microphone 30 degrees to the left and one 30 degrees to the right of the center microphone; at the condenser section, the spread of the mircophone shall be 45 degrees vice 30 degrees. Nonconformance to 3.23 shall constitute failure of this test.

4.6.3.20 Safety evaluation. Evaluation shall be accomplished throughout testing to determine conformance to MIL-STD-882. Nonconformance to 3.9 shall constitute failure of this test.

4.7 Inspection comparison. The Government may select air conditioners at any time during the contract production period and may subject these air conditioners to the examination and the tests marked "X" in column 4 of table III to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on units selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production air conditioners. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison

4.7.1 Inspection failure. Failure of an inspection comparison air conditioner to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the inspection

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comparison air conditioners and shall be cause for refusal by the Government to continue acceptance of production air conditioners until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on air conditioners previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all air conditioners accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.8 Inspection of packaging.

4.8.1 Quality conformance inspection of pack.

4.8.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.8.1.2 Lot. A lot shall consist of all completed packs offered for examination at one time.

4.8.1.3 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.8.1.4 Examination. Samples selected in accordance with 4.8.1.3 shall be examined for the characteristics listed in table V. AQL shall 2.5 percent defective.

TABLE V. Examination schedule.

Number	Characteristics	Requirement Paragraph
Major		
118.	Materials, methods, and containers not as specified. Each incorrect material, method, or container shall be considered one defect.	5.
119.	Refrigeration system not pumped down and tagged.	5.1.1.1
120.	Unit not secured to container base.	5.1.1, 5.2.1
121.	Technical publications, lifting attachments, and power connector not secured as specified.	5.1.1, 5.1.2
122.	Blocking and cushioning not as specified.	5.2.1, 5.2.2, 5.2.3
123.	Special marking not as specified.	5.3.3
124.	Marking missing, incorrect, illegible, or incomplete.	5.3

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5. PACKAGING

5.1 Preservation. Preservation shall be level A, level B, or commercial as specified (see 6.2).

5.1.1 Level A. After performing the operation specified in 5.1.1.1 the air conditioner shall be preserved without disassembly in accordance with method IIa of MIL-P-116. The air conditioner shall be secured with bolts through its base to the bottom of a box conforming to PPP-B-601, style A. The box shall be modified with two nominal 3- by 4-inch skids on the bottom attached flat and parallel to the short faces of the box. The skids shall be positioned so that the bolts securing the unit to the box bottom will pass through the skids. Bolt heads shall be countersunk in the skids. The box bottom shall be large enough to provide a minimum 1-inch clearance between the method II barrier and the faces of the box. Prior to application of the method II barriers, manuals preserved in polyethylene envelope, lifting eyes, consolidated together, and the electrical power connector placed in a polyethylene bag, shall be secured to the air conditioner housing.

5.1.1.1 Refrigeration system. The refrigeration system shall be pumped down and the valves shall be closed leaving the low-pressure side of the system with a positive pressure holding charge. A cloth tag shall be attached to the electrical power connection of the unit indicating: "Open the valves before operating the air conditoner."

5.1.2 Level B. After the refrigeration system has been pumped down and tagged in accordance with 5.1.1.1, the air conditioner shall be shrouded with polyethylene film secured in place with tape. Consolidated lifting eyes, technical publications, and power connector shall be secured to the air conditioner beneath the polyethylene shroud.

5.1.3 Commercial. Commercial preservation shall be in accordance with MIL-STD-1188. The refrigeration system shall be pumped down and tagged as specified for level A.

5.2 Packing. Packing shall be level A, level B, or commercial as specified (see 6.2).

5.2.1 Level A. The side, end, and top panels of a box conforming to PPP-B-601, style A, shall be secured to the box bottom to which the preserved air conditioner is attached. A single intermediate vertical cleat shall be used on the exterior of each side panel for reinforcement. On the interior of the side panel which is place adjacent to the front panel of the air conditioner, a nominal 2 by 4-inch horizontal cleat with a length equal to the width of the air conditioner shall be positioned and secured so that it is directly opposite the joint between the evaporator section and the condenser fan section. Care must be used to insure that the cleat does not contact the method II barrier at the edge of the filter housing of the evaporator section. Fiberboard corner pads in accordance with MIL-STD-1186 shall be placed on all top edges of the unit. In addition, flat fiberboard pads shall be secured to all faces of the unit at the joint between the evaporator section and the condenser fan section. All corner

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pads and flat pads shall have a minimum width of 4 inches and a thickness sufficient to provide snug fit between the preserved unit and the interior box faces. The box shall be reinforced with two straps conforming to QQ-S-781, class 1, type I or IV, not less than 3/4 by 0.020 inch. Unless otherwise specified (see 6.2), strapping shall be finish B. When specified (see 6.2), strapping shall be finish A.

5.2.2 Level B. The preserved air conditioner shall be secured to a 3/4-inch thick plywood base with bolts. Nominal 3 by 4-inch skids shall be secured to the base parallel to the short face of the box and positioned so that the bolts securing the unit pass through them. The base size shall be such that a minimum 1-inch clearance will be provided between the unit and the faces of the box. Wood battens, 1-inch thick, shall be secured to the upper surface of the base and located 1/2-inch from and parallel with edge of the base. Fiberboard corner pads and flat pads shall be placed on the unit in the same location and of the same width as specified for level A. The pads shall have sufficient thickness to provide a snug fit between the preserved unit and the box faces. The box shall conform to PPP-B-640, class 2, grade A, style E except that bottom flaps shall not be required. The cover shall rest on the plywood base and shall be secured to the wood battens on the base with staples. The box shall be strapped with two straps conforming to QQ-S-781, class 1, type I or IV, finish A, size as applicable. The straps shall be applied parallel to the skids and adjacent to the inside edges of the skids.

5.2.3 Commercial. Each air conditioner, complete with technical publications, lifting eyes, and power connector shall be packed in accordance with MIL-STD-1188 (see 6.7).

5.3 Marking.

5.3.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

5.3.2 Commercial packaging. Commercial marking shall be in accordance with MIL-STD-1188.

5.3.3 Special marking. Special markings shall be placed on two sides of the container for all degrees of packaging to indicate that the air conditioner shall remain in operation position during shipment and storage. Markings shall consist of arrows and words "THIS SIDE UP" in black, 2-inch-high letters placed in the upper third of the container height.

6. NOTES

6.1 Intended use. The air conditioners covered by this specification are intended for use in cooling, dehumidifying, filtering, and circulating air for portable shelters or vans in accordance with electronic and personnel requirements. The air conditioners may be installed indoors or directly exposed to the weather.

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6.2 Ordering data. Proceurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Electrical voltage required (see 3.15).
(c) Time frame required for submission of preproduction model and number required (see 3.2).
(d) When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some but not all of the preproduction examination and tests, the contracting officer should specify which examination and test will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
(e) When initial production inspection is required and number of air conditioners to be furnished when applicable (see 3.3).
(f) When control panel is to be mounted on an end panel of the evaporator blower section. Specify left or right side as viewed from the front of the air conditioner (see 3.13).
(g) When color of finish coat is other than as specified (see 3.19).
(h) Degrees of preservation and of packing required (see 5.1 and 5.2).
(i) When other than finish B strapping is required (see 5.2.1).

6.3 Preproduction model. Any changes or deviations of production air conditioners from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish air conditioners conforming to this specification.

6.4 Definitions.

6.4.1 End item final inspection requirement (EIFIR). A EIFIR is a check list of quality characteristics prepared by the Government which the contractor must follow during his final inspection of the completed end item to verify complete and functional conformance to contract requirements prior to submission for final acceptance.

6.5 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operational manual to be furnished with each air conditioner.

6.6 Provisioning. The contracting officer should include provisioning requirements for repair parts as necessary and instructions on shipment of air conditioners.

6.7 Packing. Air conditioners packed commercially should be in containers unless otherwise specified. Containers are required to provide stacking capabilities and facilitate handling during shipment and storage.

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6.8 Electromagnetic interference.

(a) This air conditioner has been previously procured and qualified to MIL-STD-461A, Notice 1, and MIL-STD-462, Notice 1, as specified in Sections 3 and 4 of this specification. Susequent notices and revisions should not be made applicable to a procurement contract.

(b) The contracting officer should include the following in procurement documents:

(1) Notices and revisions other than those listed below are not applicable.

MIL-STD-461A, Notice 1.
MIL-STD-462, Notice 1.

(2) Bidders are cautioned against failing to properly take into consideration the Radio Frequency Interference Suppression/ Electromagnetic Interference Characteristics Requirements for the end items in preparing their bids.

The foregoing are performance requirements only. Specific drawings which may be incorporated therein are intended as an example of a method which was successful in the past. Since each offeror has latitude in selecting electrical components within the requirements of the drawings and specifications, and the electron-magnetic emission and suppression characteristics may differ from those previously used, the Government does not expressly or impliedly warrent that performance requirements will be met if such drawings are followed and reliance upon such drawings is at the contractor's own risk.

6.9 Finish paint coat validation. Prior to acceptance of any finish coat, the contracting officer should require validation of spectral reflectance characteristics of the finish coat from the Organic and Chemical Coating Research Team, Material Technology Laboaratory, DRDME-VO, USA Mobility Equip-ment Research and Development Command, Fort Belvoir, VA 22060.

6.10 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.4).

Custodian:
Army - ME

Review activities:
DLA - GS

User activities:
Army - AT, MI, CR
Navy - MC

Preparing activity:
Army - ME

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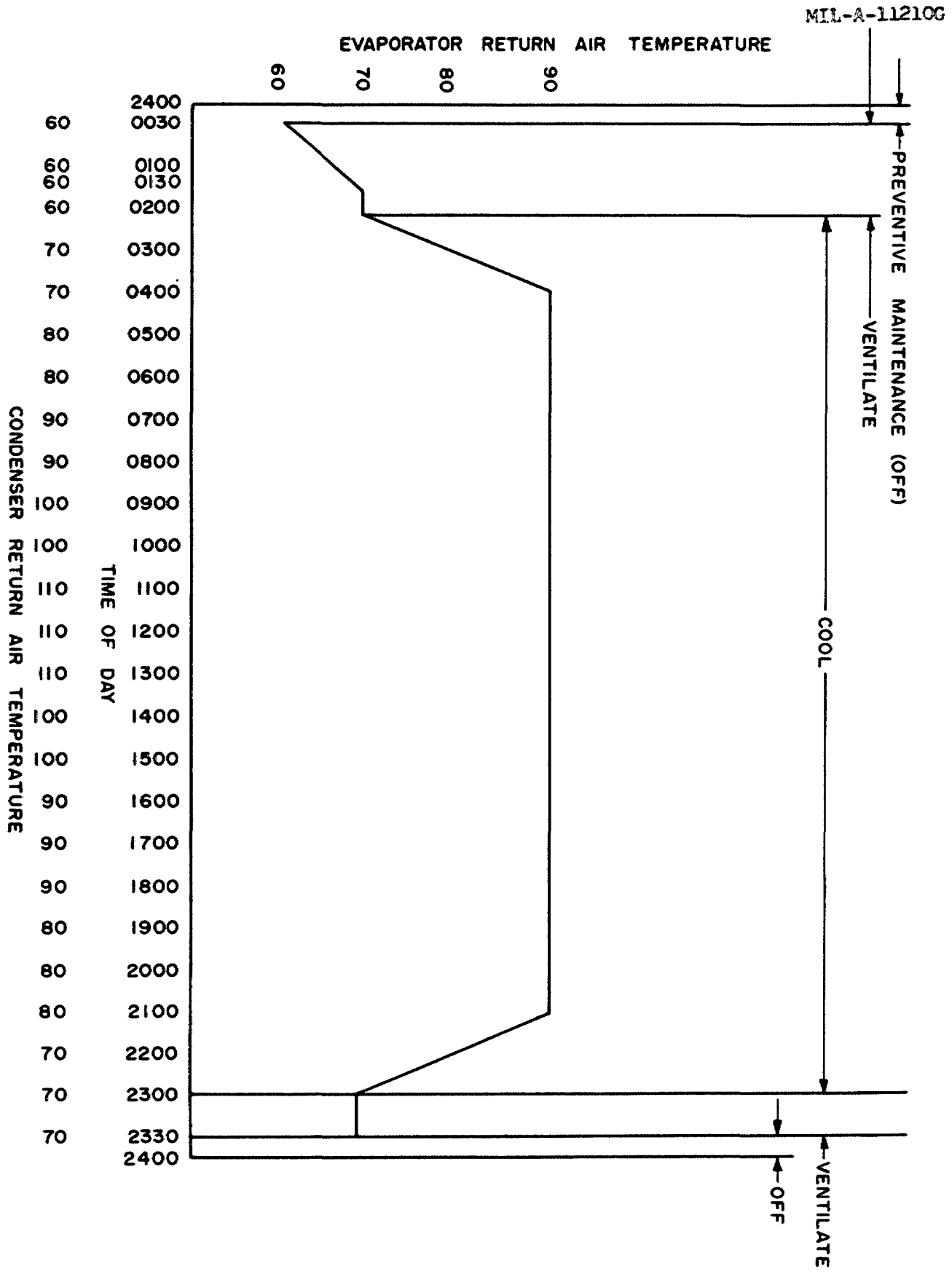
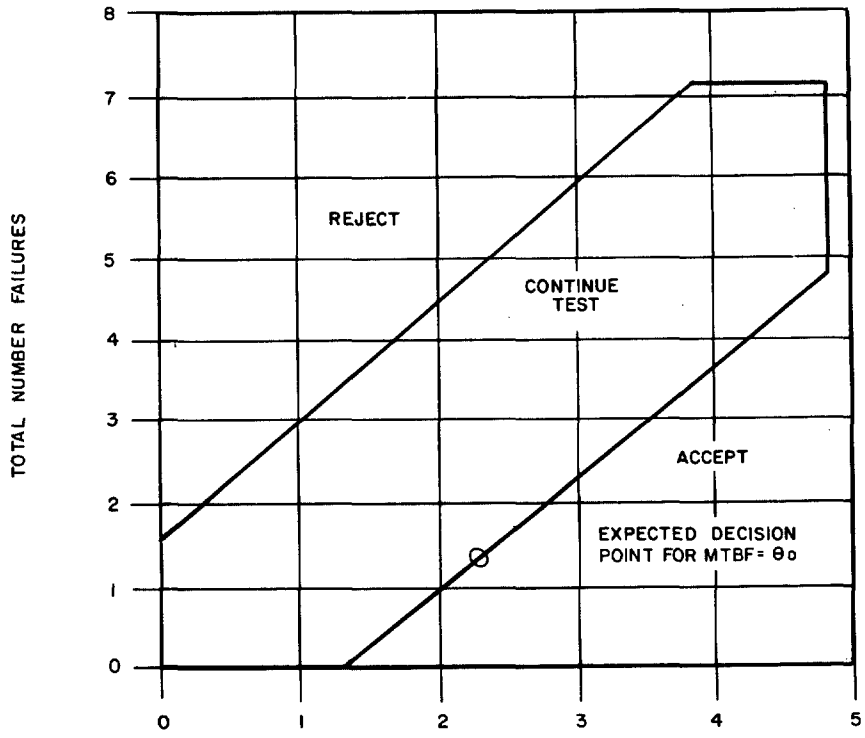


Figure I. AIR CONDITIONER TEMPERATURE - TIME TEST PROFILE

X-2331

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DECISION RISKS 20%
DISCRIMINATION RATIO 2.0:1



NO. OF FAILURES	TOTAL TEST TIME * REJECT (EQUAL OR LESS)	ACCEPT EQUAL OR MORE
0	N/A	1.40
1	N/A	2.09
2	.35	2.79
3	1.04	3.48
4	1.73	4.17
5	2.43	4.87
6	3.12	4.87
7	3.81	4.87
8	4.87	N/A

*TOTAL TEST TIME IS TOTAL UNIT HOURS OF "EQUIPMENT ON" TIME AND IS EXPRESSED IN MULTIPLES OF THE SPECIFIED MTBF.

FIGURE 2. Accept-reject criteria.

X-3661

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-A-11210G Air Conditioners, Base Mounted, Self-Contained, Multipackage-Type, Air-Cooled, EMD, 18,000-BTU/Hr

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

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